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Stabilization and Regression of Albuminuria in Chinese Patients With Type 2 Diabetes: A One-year Randomized Study of Valsartan Versus Enalapril

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ABSTRACT

This study was designed to compare the short-term (1-y) tolerability and antiproteinuric efficacy of enalapril and valsartan in patients with type 2 diabetes. Forty-two patients with normal renal function or early-stage nephropathy were recruited in Hong Kong and randomized to valsartan 80 mg/day or enalapril 5 mg/day; the doses were increased to 160 mg and 10 mg daily, respectively, as tolerated. Early-morning urine was analyzed for albumin and creatinine and 24-hour urinary albumin excretion at baseline and 1 year after therapy began. Twenty-two patients were randomized to valsartan and 20 to enalapril. The 2 treatment groups were similar in terms of age, sex distribution, and duration of diabetes or hypertension. Blood pressure decreased to a similar extent (–2.5% to –5.0%) with each drug. Similarly, the 24-hour urinary albumin excretion decreased by 5% to 6% with each drug. The albumin-creatinine ratio in early-morning urine samples and plasma creatinine levels decreased in the valsartan group and increased in the enalapril group, but the difference was not significant. Plasma potassium levels were stable in both groups at the end of study. Cough was reported by 7 (35%) patients receiving enalapril and none of those receiving valsartan (P=.003). In conclusion, enalapril and valsartan both reduced blood pressure and albuminuria to a similar extent with 1 year of therapy in Chinese patients with type 2 diabetes and normal renal function or early-stage nephropathy. Fewer adverse events were reported with valsartan, but both drugs appear to be relatively safe.

Keywords: valsartan; enalapril; type 2 diabetes; Chinese; diabetic nephropathy; albuminuria

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INTRODUCTION

Nephropathy is one of the major microangiopathic complications of diabetes. In Hong Kong, type 2 diabetes mellitus is the leading underlying cause of renal failure resulting in the need for transplantation among Chinese patients.[1,2] Diabetic nephropathy results in persistent albuminuria, which can progress to renal failure.[3,4] Comorbid hypertension results in a strong risk factor for albuminuria and nephropathy.[5,6]

To advance understanding of the role of the renin-angiotensin system (RAS) in the development of nephropathy, clinical researchers have extensively investigated medications acting on RAS. Both angiotensin-converting enzyme (ACE) inhibitors and angiotensin II receptor blockers (ARBs) are able to modulate RAS and have been proven to exert beneficial effects on blood pressure (BP) and urinary albumin excretion (UAE).[7,8] Very few studies have directly compared the effects of an ACE inhibitor with those of an ARB in type 2 diabetes, however. Information on their effects in Chinese patients is even more scanty. In this 1-year randomized trial, we compared the effects of enalapril (Renitec™*), an ACE inhibitor, with those of valsartan (Diovan™†), an ARB, on BP and albuminuria in Chinese patients with type 2 diabetes and normal renal function or early-stage nephropathy. The purpose of this study was to compare the efficacy and tolerability of enalapril versus valsartan in Chinese patients with diabetes.

PATIENTS AND METHODS

A total of 42 Chinese patients with type 2 diabetes and normal renal function or early-stage nephropathy were recruited for this study. The study involved a 2-week run-in baseline period and 1 year of active treatment. Patients enrolled in this study were randomized to treatment with valsartan 80 mg daily or enalapril 5 mg daily. The randomization process involved the use of sealed envelopes. Doses were titrated to 160 mg and 10 mg for valsartan and enalapril, respectively, provided the systolic BP remained >110 mm Hg and the drug was well tolerated. The target BP was <140/90 mm Hg. Patients were allowed to continue other drugs (eg, antidiabetic drugs) at the same dosage as before they enrolled in the study.

Chinese patients with type 2 diabetes aged 30 to 80 years were eligible. Patients were excluded if they had uncontrolled hypertension (sitting BP >200/115 mm Hg); a history of myocardial infarction, cerebrovascular accident, or uncontrolled congestive heart failure within the previous 6 months; or significant renal impairment (plasma creatinine ≥150 µmol/L).

During the baseline period (week 2), BP, body mass index (BMI), renal and liver functions, and glycemic indices (glycated hemoglobin [HbA_{1c}], fasting plasma glucose [PG]) were measured, and early-morning urine was collected to determine the albumin-creatinine ratio (ACR) and 24-hour UAE. At the end of the baseline period (week 0), patients who met all inclusion criteria and none of the exclusion criteria were randomized to active treatment. After commencing valsartan or enalapril therapy, all patients returned 2 weeks later for an assessment of renal function. They were then monitored every 3 months to evaluate tolerance of the study medications, BP, and renal function. They were also assessed at week 52 for glycemic indices, ACR, and 24-hour UAE.

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*Renitec (enalapril maleate, MSD) is a registered trademark of Merck & Co., Inc., Whitehouse Station, NJ.
†Diovan (valsartan) is a registered trademark of Novartis Pharmaceutical Corp., East Hanover, NJ.

Urinary albumin levels were considered acceptable (normoalbuminuria) when the 24-hour UAE was <30 mg/d and spot urinary ACR readings were <3 mg/mmol. Microalbuminuria was defined by a 24-hour UAE of 30 to 300 mg/d or spot urinary ACR readings of 3 to 30 mg/mmol; macroalbuminuria was defined by a 24-hour UAE of 300 mg/d or spot urinary ACR readings of ≥30 mg/mmol. Regression of albuminuria was defined by the conversion of macroalbuminuria to microalbuminuria or normoalbuminuria or the conversion of microalbuminuria to normoalbuminuria. Stabilization was defined by a microalbuminuria or normoalbuminuria that remained static throughout the therapeutic period.

All patients were ethnic Chinese recruited from the Department of Medicine of Alice Ho Miu Ling Nethersole Hospital in Hong Kong. The study protocol was approved by the ethical committee of the hospital.

Analytical Methods

PG was measured by the hexokinase method (Hitachi 911 Analyzer, Boehringer Mannheim, Mannheim, Germany). The intra-assay and interassay coefficients of variation for glucose were 2% at 6.6 mmol/L. HbA_{1c} was measured by automated ion-exchange chromatography (Bio-Rad Laboratory, Hercules, Calif, USA; manufacturer's reference range, 5.1%–6.4%). The intra-assay and interassay coefficient of variation for HbA_{1c} was 3.1% at values below 8.5%.

Sample Size Estimation

With the clinically relevant difference in 24-hour urine albumin expected to be 10 mg/d after treatment and the standard deviation 7.5 mg/d, the standardized difference would be 1.3. The power was set at 0.8. The sample size required for clinical relevance was calculated at 20 patients in each arm.

Statistical Analysis

The analysis was carried out using data from the intention-to-treat population. It was performed using SPSS software version 10.0 on an IBM-compatible computer. Spot urine ACR and 24-hour urine albumin were logarithmically transformed because of a skewed distribution. All results are expressed as mean±SD, geometric mean multiplied or divided by antilog SD or number (percentage), where appropriate. The analysis of variance, paired *t* test, and χ^2 tests were used for between-group comparisons, where appropriate. The Fisher exact test was used for group comparisons when one or more of the groups consisted of fewer than 5 patients. A *P* value <.05 was considered significant.

RESULTS

Of the 42 patients enrolled, 17 (40.5%) were men and 25 (59.5%) were women. The mean ±SD age was 61.0±11.1 years (median, 62.5 y; range, 31–80 y). Twenty-two patients were randomized to valsartan and 20 to enalapril. All patients had hypertension and were already taking antihypertensive agents at the time of enrollment. None of the patients reported taking an ACE inhibitor or ARB. The 2 treatment groups were similar in terms of age, sex distribution, and duration of diabetes or hypertension (Table 1). The mean dosages used throughout the study were 6.3±2.2 mg for enalapril

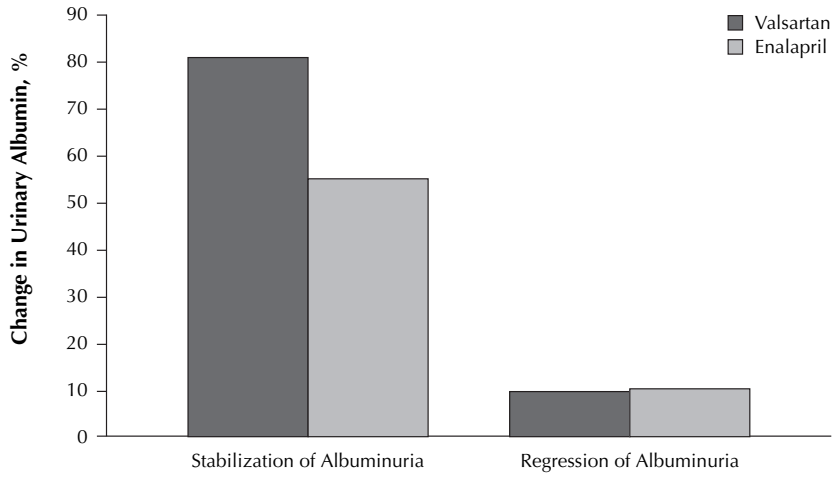
and 109.1±39.4 mg for valsartan. Five (25%) patients used enalapril 10 mg and 8 (36.4%) used valsartan 160 mg.

In each treatment group, BP decreased by 2.5% to 5.0%, and the reduction was similar between the 2 groups (Table 1). The 24-hour urine showed a comparable reduction in albumin—5% to 6% after valsartan or enalapril treatment, respectively—but the difference was not significant. Urinary ACR and plasma creatinine levels de-creased in the valsartan group and increased in the enalapril group, but the difference was not significant. Plasma potassium levels remained stable in both treatment groups until study end.

Only 1 patient (from the valsartan group) dropped out before the end of the study (at week 36). Seven (35%) patients receiving enalapril and none of those receiving valsartan complained of cough (*P*=.003). More patients in the enalapril group (n=9) reported adverse events (AEs) during the treatment period compared with those in the valsartan group (n=3; 45.0% vs 13.6%; *P*=.015). AEs included finger numbness (n=1, valsartan group), joint pain (n=1, valsartan group), palpitations (n=1, enalapril group), minor stroke (n=1, enalapril group), and cough (n=7, enalapril group). However, none of the study participants dropped out early because of treatment-related side effects.

After 1 year of therapy, urinary albumin levels remained stable in 17 (81%) patients receiving valsartan and 11 (55%) patients receiving enalapril (Fig). The difference fell just short of statistical significance (*P*=.073). Regression was observed in 2 patients in each group (9.5% vs 10%, respectively; *P*=.678). Table 2 summarizes changes in the status of albuminuria in all patients after 1 year of therapy.

Stabilization and regression of albuminuria after 1-year of treatment with valsartan versus enalapril.



P values for valsartan vs enalapril: stabilization of albuminuria, *P*=.073; regression of albuminuria, *P*=.678.

Table 1. Clinical Characteristics of the Study Participants*

Characteristic	Overall (n=42)	Valsartan (n=22)	Enalapril (n=20)	*P* Value
Baseline				
Age, y	61.0±11.1	62.1±10.5	59.7±11.8	.484
Men, n (%)	17 (40.5)	9 (40.9)	8 (40.0)	.954
Duration of diabetes, y	9.6±6.1	9.2±5.5	10.1±6.8	.651
Duration of hypertension, y	6.6±5.1	6.8±5.2	6.5±5.1	.841
BMI, kg/m^2	25.3±2.8	25.1±3.0	25.6±2.7	.573
Systolic BP, mm Hg	143±17	144±20	142±13	.703
Diastolic BP, mm Hg	77±10	79±8	76±11	.301
Fasting PG, mmol/L	8.5±2.5	8.6±2.5	8.3±2.4	.683
HbA_{1c}, %	7.6±1.7	8.0±1.7	7.2±1.8	.145
Creatinine, µmol/mL	84±32	86±30	83±37	.774
Potassium, mmol/L	4.4±0.5	4.4±0.5	4.5±0.5	.462
Spot urinary ACR, mg/mmol†	5.1 ×/÷ 6.6	3.7 ×/÷ 6.5	7.1 ×/÷ 6.7	.405
24-hour urinary albumin, mg/d	70.4 ×/÷ 7.5	40.1 ×/÷ 7.0	114.0 ×/÷ 7.6	.202
End of study				
BMI, kg/m^2	25.5±2.6	25.3±2.7	25.7±2.5	.594
Systolic BP, mm Hg	137±16	139±17	136±16	.509
Diastolic BP, mm Hg	76±8	77±10	74±6‡	.280
Fasting PG, mmol/L	8.1±2.7	8.6±3.0	7.7±2.3	.289
HbA_{1c}, %	7.6±1.3	7.8±1.4	7.5±1.3	.451
Creatinine, µmol/mL	90±48	83±34	97±58	.343
Potassium, mmol/L	4.5±0.6	4.5±0.4	4.6±0.7	.469
Spot urinary ACR, mg/mmol†	7.5 ×/÷ 7.2	4.6 ×/÷ 6.6	12.8 ×/÷ 7.4	.161
24-hour urinary albumin, mg/d†	58.0 ×/÷ 8.3	39.3 ×/÷ 6.6	83.9 ×/÷ 9.4	.270
Percentage change				
BMI, %	1.1±3.7	1.4±3.2	0.8±4.2	.657
Systolic BP, %	–3.4±11.4	–2.5±12.5	–4.3±10.4	.619
Diastolic BP, %	–1.2±12.6	–2.5±13.7	–5.0±10.3	.054
Fasting PG, %	–1.4±31.8	–0.1±25.8	–2.6±37.6	.804
HbA_{1c}, %	7.9±56.6	–1.5±13.5	17.7±79.7	.284
Creatinine, %	25.4±142.6	–3.4±15.2	55.5±201.8	.190
Potassium, %	2.9±11.6	2.5±10.7	3.4±12.7	.822
Spot urinary ACR, %	13±162	–8±131	34±192	.453
24-hour urinary albumin, %	–5±27	–6±11	–5±36	.906

*All data expressed as value ±SD except where indicated.
†Geometric mean ×/÷ antilog SD paired *t* test comparing baseline and end of study.
‡*P* values for enalapril, baseline vs final visit <.05.

Table 2. Status of Albuminuria in Study Participants Before and 1 Year After Beginning Treatment

Albuminuria	Treatment	Baseline, n (%) (n=42)	End of Study, n (%) (n=41)	*P* Value*
Normoalbuminuria†	Valsartan	9 (40.9)	10 (47.6)	.473
	Enalapril	6 (30.0)	5 (25.0)	.140
Microalbuminuria	Valsartan	10 (45.5)	8 (38.1)	.977
	Enalapril	9 (45.0)	9 (45.0)	.663
Macroalbuminuria	Valsartan	3 (13.6)	3 (14.3)	.361
	Enalapril	5 (25.0)	6 (30.0)	.235
Micro- or macroalbuminuria	Valsartan	13 (59.1)	11 (52.4)	.473
	Enalapril	14 (70.0)	15 (75.0)	.140

**P* values for valsartan vs enalapril: all insignificant.
†Normoalbuminuria was defined by UAE <30 mg/d and spot urinary ACR <3 mg/mmol.

DISCUSSION

Albuminuria is common in Chinese patients with type 2 diabetes in Hong Kong. MAPS (Microalbuminuria Prevalence Study), conducted in hypertensive patients with type 2 diabetes in Hong Kong, showed that the prevalence of microalbuminuria and macroalbuminuria was 24.9% and 18.3%, respectively.[9] Albuminuria is not only a feature of diabetic nephropathy but is also associated with increased mortality and morbidity.[10,11] Increased RAS activity is one of the major pathogenetic mechanisms underlying diabetic nephropathy. Angiotensin II has major effects on renal hemodynamics and cellular growth.[12] Treatment with an ACE inhibitor is associated with a reduction in proteinuria in both type 1 and type 2 diabetes that is independent of the degree of BP reduction.[13-15] However, when renal disease progression is used as a study end point, ACE inhibitors have been proven useful in type 1 diabetes but not type 2.[16]

ARBs are effective in reducing proteinuria. The IRMA (Irbesartan in Patients With Type 2 Diabetes and Microalbuminuria) study reported that ARBs slowed the progression of albuminuria in type 2 diabetes independent of their effect on BP.[17] Other studies—RENAAL (Reduction of Endpoints in NIDDM with the Angiotensin II Antagonist Losartan), IDNT (Irbesartan Diabetic Nephropathy Trial), and MARVAL (MicroAlbuminuria Reduction with Valsartan)—also demonstrated that ARBs have a slowing effect on the progression of diabetic nephropathy in type 2 diabetes.[18-20] Preliminary data suggest that a combination of ACE inhibitor and ARB therapy might have a more potent beneficial effect on patients with type 2 diabetes.[21,22] Nevertheless, initial therapy with either an ACE inhibitor or ARB is still the mainstay of clinical practice. So far, limited information is available on direct comparisons of the antiproteinuric effect of ACE inhibitors and ARBs in type 2 diabetes.

Lacourciere et al compared losartan with enalapril in patients with type 2 diabetes and early-stage nephropathy and found that both drugs reduced UAE to a similar extent.[23] The recently published DETAIL (Diabetics Exposed to Telmisartan and Enalapril) study prospectively followed 250 patients with type 2 diabetes for 5 years after randomization to either telmisartan or enalapril.[24] They found that telmisartan was not inferior to enalapril in providing long-term renoprotection, such that the changes in glomerular filtration rate were similar in both groups.[24]

In Chinese patients with type 2 diabetes and comorbid hypertension, enalapril has been reported to be more effective than nifedipine (a calcium antagonist)[14,25] in reducing proteinuria. As far as we know, before this study there were no published data on head-to-head comparisons of ACE inhibitors and ARBs in Chinese patients with diabetes. In accord with the study by Lacourciere et al,[23] a similar improvement in 24-hour UAE in these patients with enalapril or valsartan therapy was found. More patients in the valsartan group achieved stabilization of albuminuria (81% vs 55%), but the difference was not significant (P=.073). This is consistent with previous reports that the antiproteinuric effect is stronger with ARB therapy than with ACE inhibitors in patients with nondiabetic nephropathy (eg, chronic glomerulonephritis).[26,27] On the other hand, AEs, especially cough, occurred much less frequently in patients taking an ARB compared with those taking an ACE inhibitor; however, most of the AEs were minor. Some authorities favor the use of ARBs more than ACE inhibitors in type 2 diabetes[7]; however, a consensus on this issue is still lacking.

CONCLUSION

In Chinese patients with type 2 diabetes and normal renal function or early-stage nephropathy, enalapril and valsartan demonstrated similar efficacy in terms of BP reduction and albuminuria with 1 year of therapy. Fewer AEs were reported with valsartan compared with enalapril, but both drugs are considered relatively safe.

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